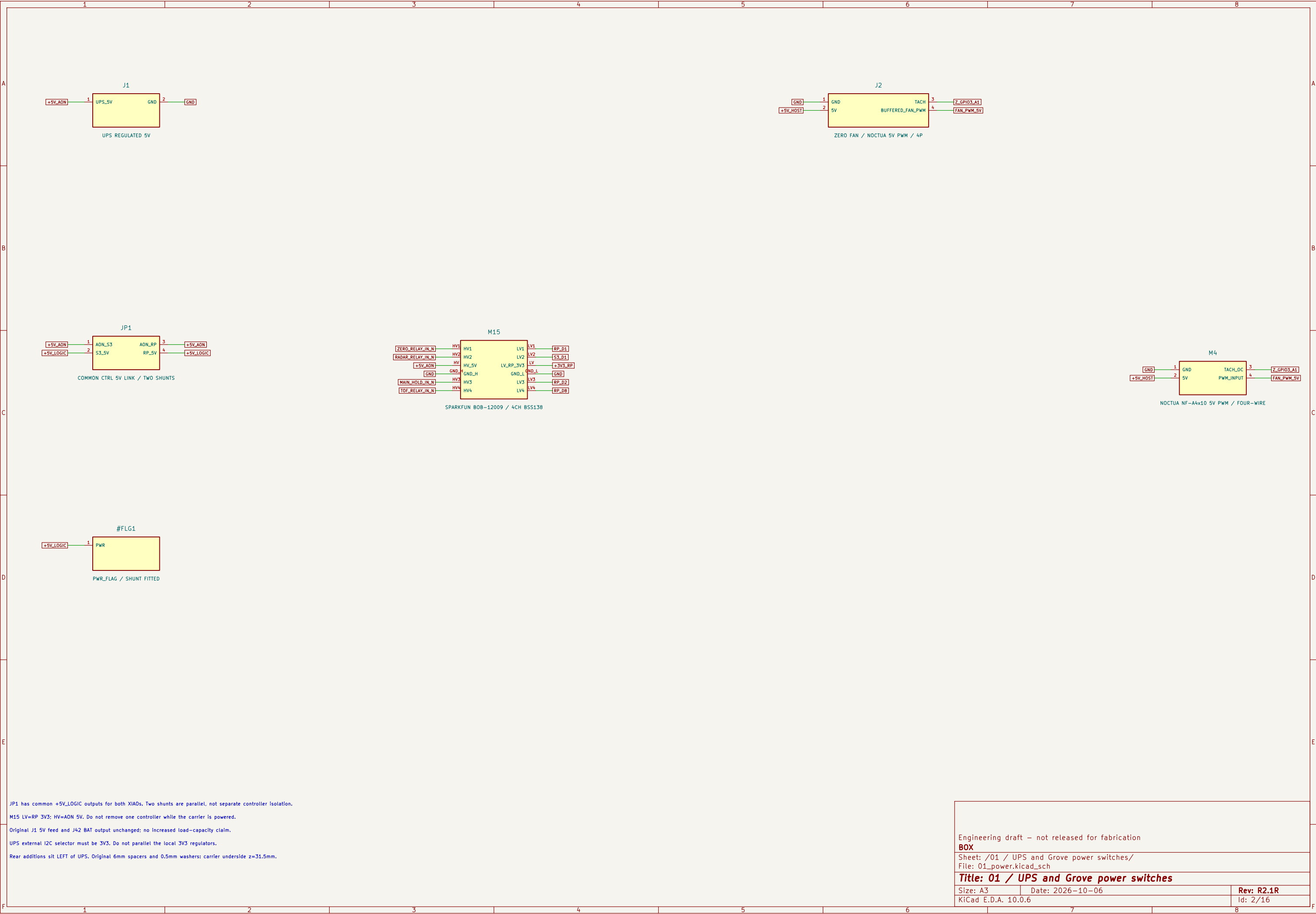


	1	2	3	4	5	6	7	8												
A	01 / UPS and Grove power switches File: 01_power.kicad_sch		02 / Finished boards and expansion headers File: 02_computers.kicad_sch		03 / Grove interfaces and module adapter ports File: 03_ports.kicad_sch															
B	08 / Wired expansion sockets File: 08_expansion_ports.kicad_sch		04 / Shutdown control and selected debug points File: 04_control_debug.kicad_sch		09 / Complete UPS connections and rear power path File: 09_ups_and_rear_power.kicad_sch															
C	07 / Front SEN0628 UART4 File: 07_front_tof.kicad_sch		05 / Complete Servo Driver HAT B module File: 05_servo_module.kicad_sch		06 / Four remote HLK-LD2450 harnesses File: 06_radar_harness.kicad_sch															
D	10 / Existing ReSpeaker USB link File: 10_respeaker_usb.kicad_sch		11 / Existing EYES, IMU and HAT connections File: 11_verified_signals.kicad_sch		12 / XIAO original edge GPIO breakout File: 12_xiao_extra_gpio.kicad_sch															
E	12 / Three load relays and whole-device power hold File: 12_four_power_relays.kicad_sch		13 / Rear complete UART modules File: 13_power_off_uart.kicad_sch		14 / Rear fan PWM complete module File: 14_fan_buffer.kicad_sch															
F	ENGINEERING: module fit, contact load and physical qualification remain open. See current README.					<table><tr><td colspan="3">Sheet: / File: box_carrier_modules_r2.kicad_sch</td></tr><tr><td colspan="3">Title: BOX module carrier R2 – engineering draft</td></tr><tr><td>Size: A3</td><td>Date: 2026-10-06</td><td>Rev: R2.1R</td></tr><tr><td colspan="2">KiCad E.D.A. 10.0.6</td><td>Id: 1/16</td></tr></table>			Sheet: / File: box_carrier_modules_r2.kicad_sch			Title: BOX module carrier R2 – engineering draft			Size: A3	Date: 2026-10-06	Rev: R2.1R	KiCad E.D.A. 10.0.6		Id: 1/16
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Size: A3	Date: 2026-10-06	Rev: R2.1R																		
KiCad E.D.A. 10.0.6		Id: 1/16																		
	1	2	3	4	5	6	7	8												



JP1 has common +5V_LOGIC outputs for both XIAOs. Two shunts are parallel, not separate controller isolation.

M15 LV=RP 3V3; HV=AON 5V. Do not remove one controller while the carrier is powered.

Original J1 5V feed and J42 BAT output unchanged; no increased load-capacity claim.

UPS external I2C selector must be 3V3. Do not parallel the local 3V3 regulators.

Rear additions sit LEFT of UPS. Original 6mm spacers and 0.5mm washers: carrier underside z=31.5mm.

Engineering draft – not released for fabrication

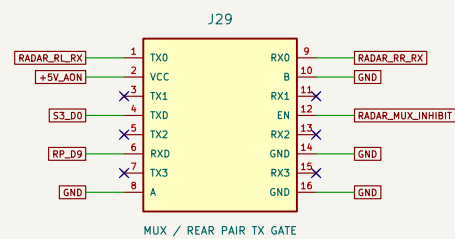
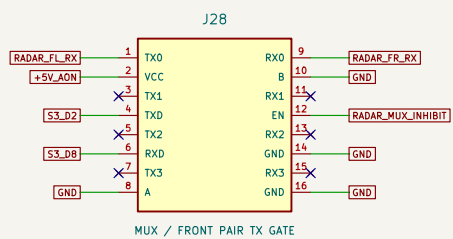
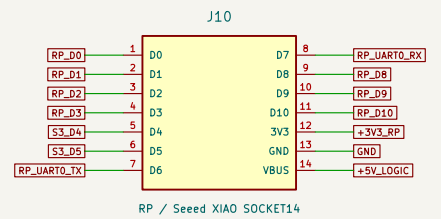
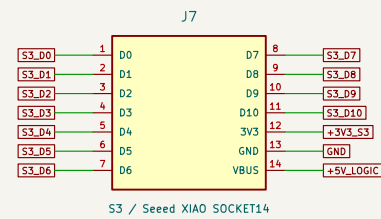
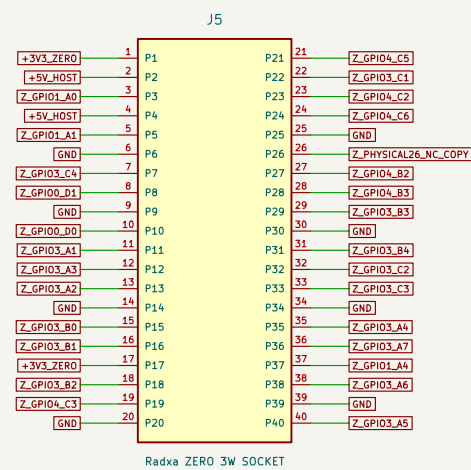
BOX

Sheet: /01 / UPS and Grove power switches/
File: 01_power.kicad_sch

Title: 01 / UPS and Grove power switches

Size: A3Date: 2026-10-06KiCad E.D.A. 10.0.6

Rev: R2.1RId: 2/16



XIAO modules plug into fourteen edge contacts. Bottom pads and fine-pitch interfaces are not silently connected.

J33 copies all 40 physical ZERO contacts for removable HAT/general expansion; no extra J30 strip.

ZERO socket pin-1/model registration and external USB VBUS blocking require final qualification.

S3 GPIO43/44 (D6/D7) no longer connect to a radar: ROM UART is kept off sensor RX.

Radar RX Inputs on S3 GPIO4/8/9 and RP GPIO5: Input only, disable all pull-ups.

Both CD4052 VCC pins use AON 5V. EN must be active-HIGH INHIBIT at module pin 12.

M2 OFF -> NC supplies AON 5V to both EN pins. ON -> NC open, module pulldowns enable.

Before M2 ON set four TX pins high-Z; wait for radar power to settle, then start UART.

RX lines are not galvanically isolated. Common-supply collapse/brownout requires bench validation.

MUX EN is not connected to M15 HV2; relay control cannot be loaded by EN pulldowns.

Engineering draft – not released for fabrication

BOX

Sheet: /02 / Finished boards and expansion headers/

File: 02_computers.kicad_sch

Title: 02 / Finished boards and expansion headers

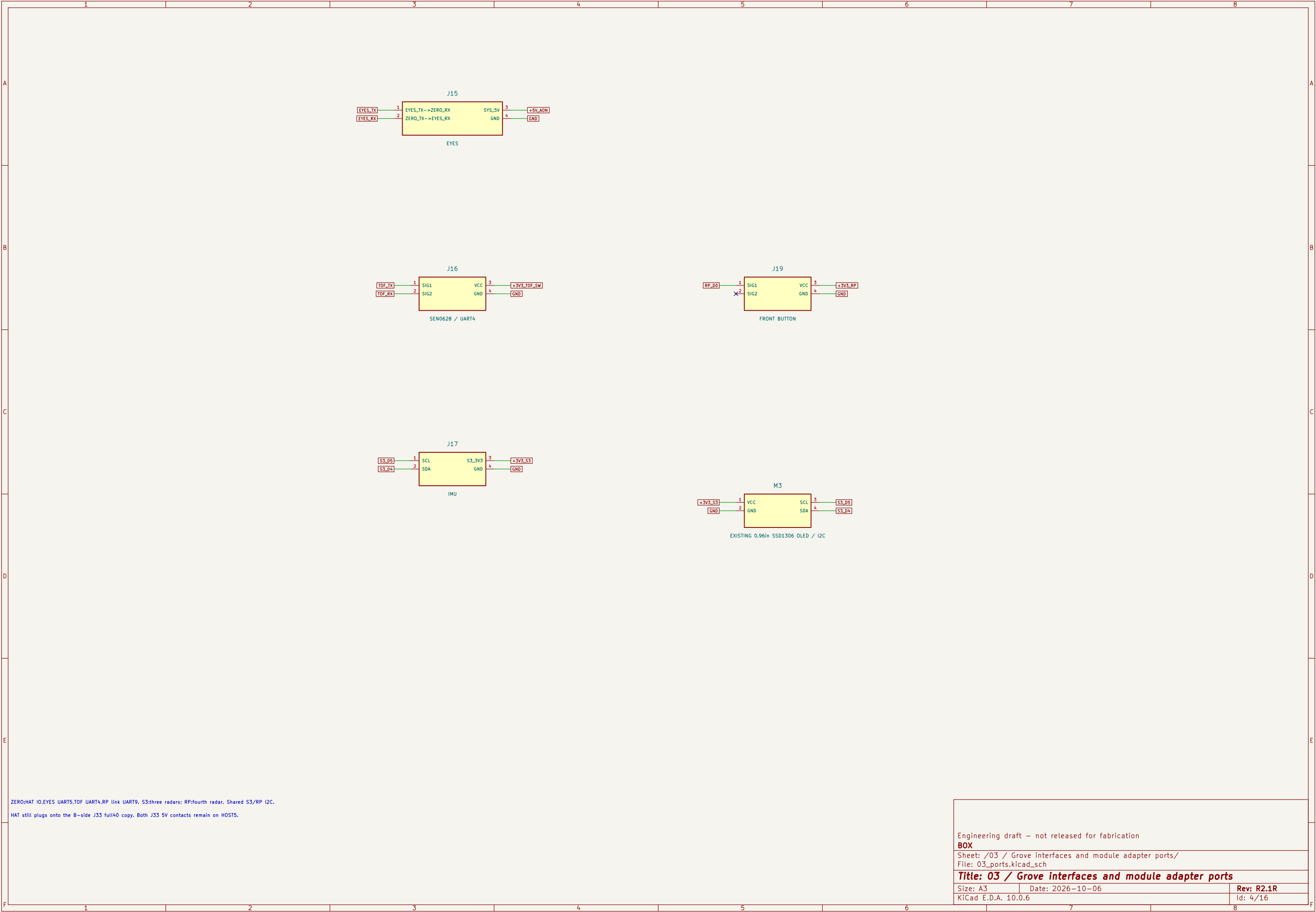
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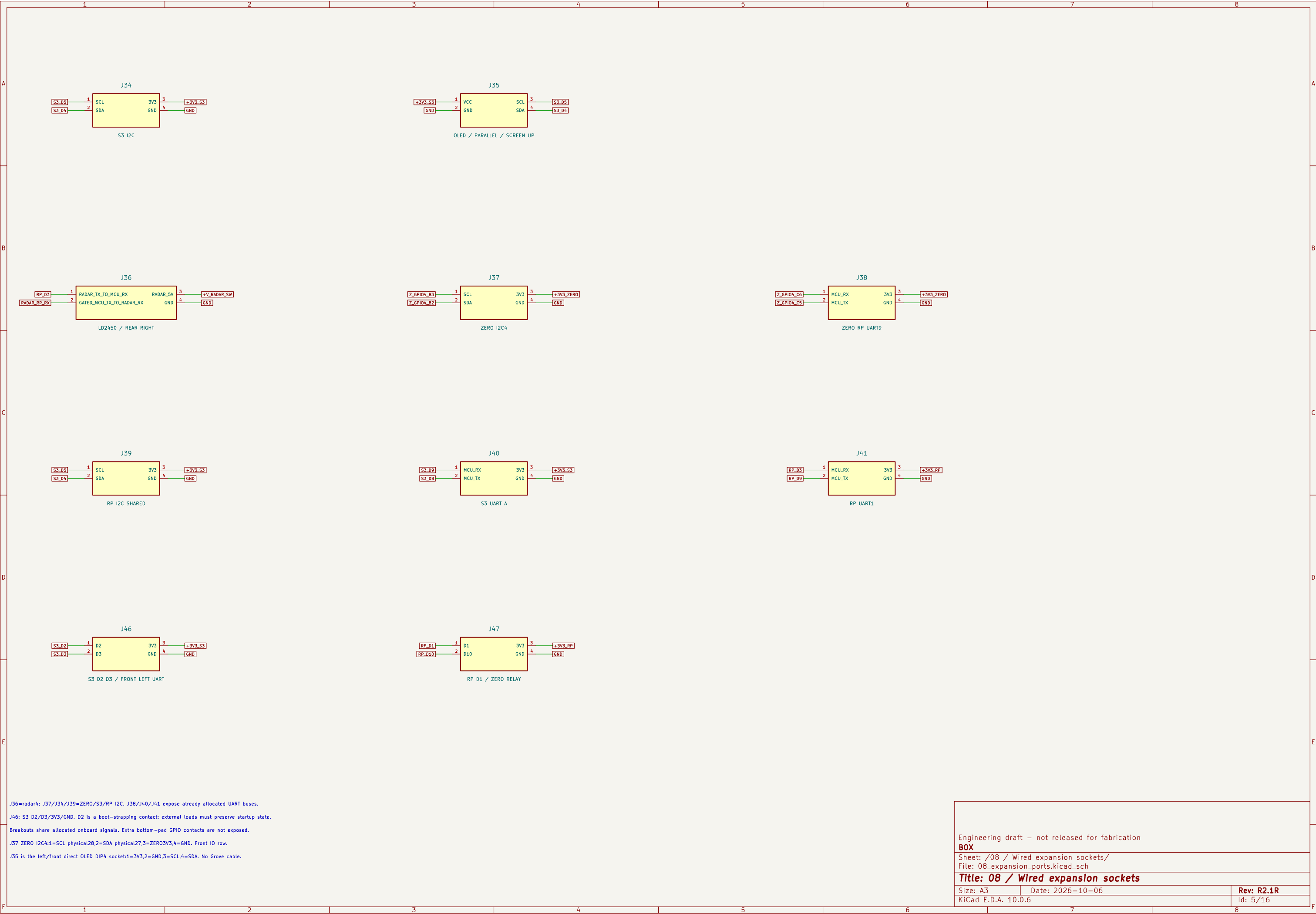
Date: 2026-10-06

Rev: R2.1R

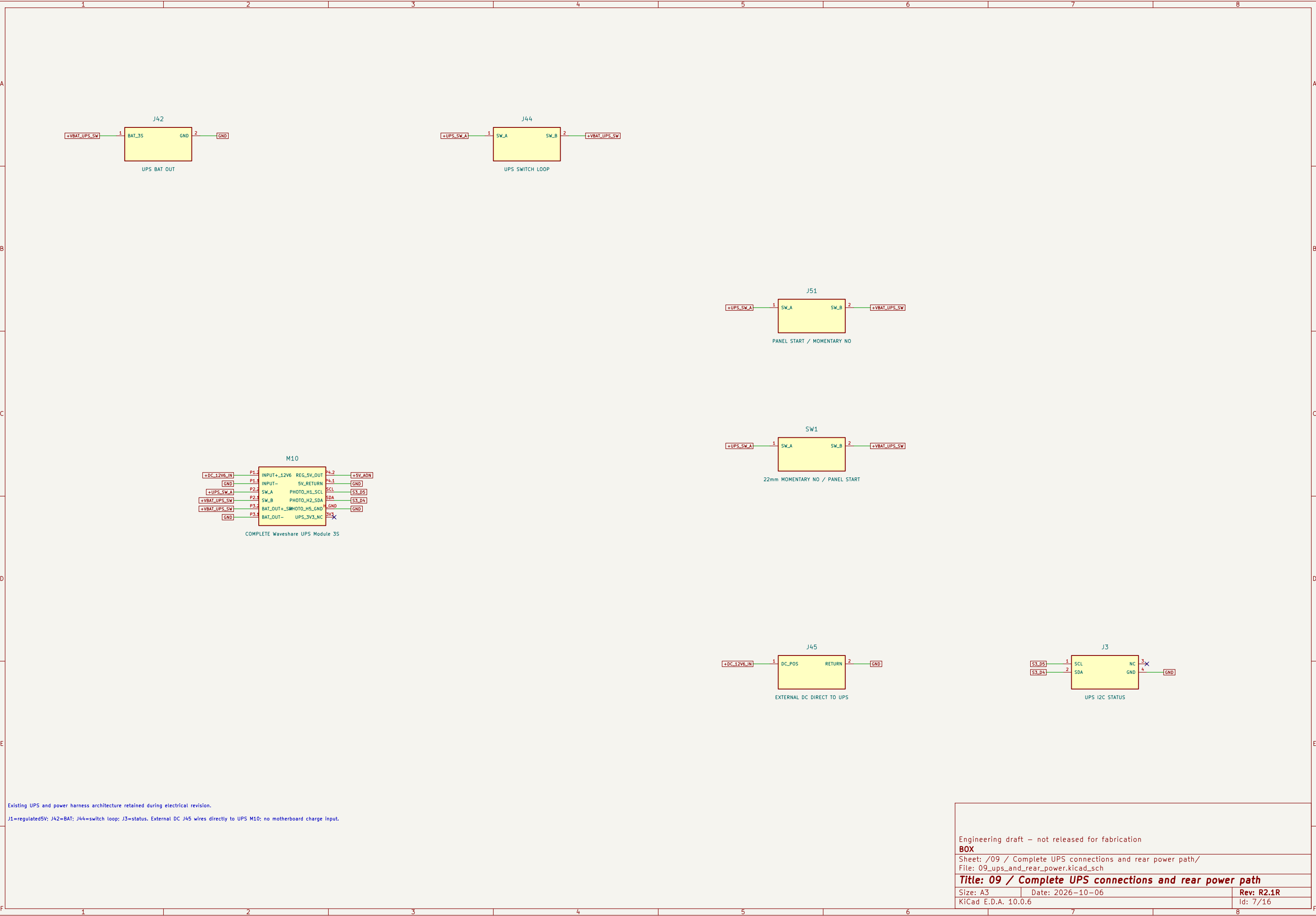
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Id: 3/16











A

B

C

D

E

F

A

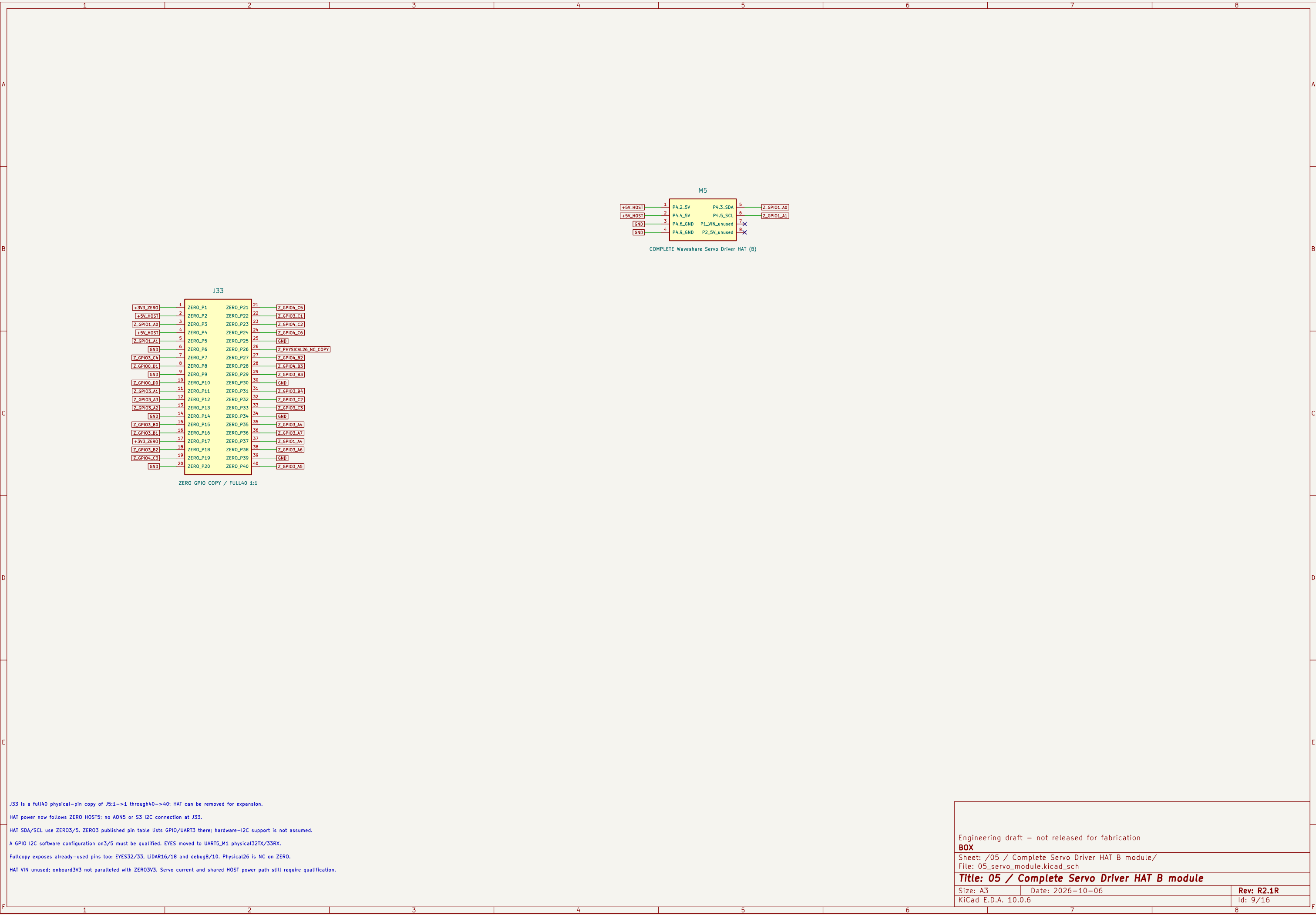
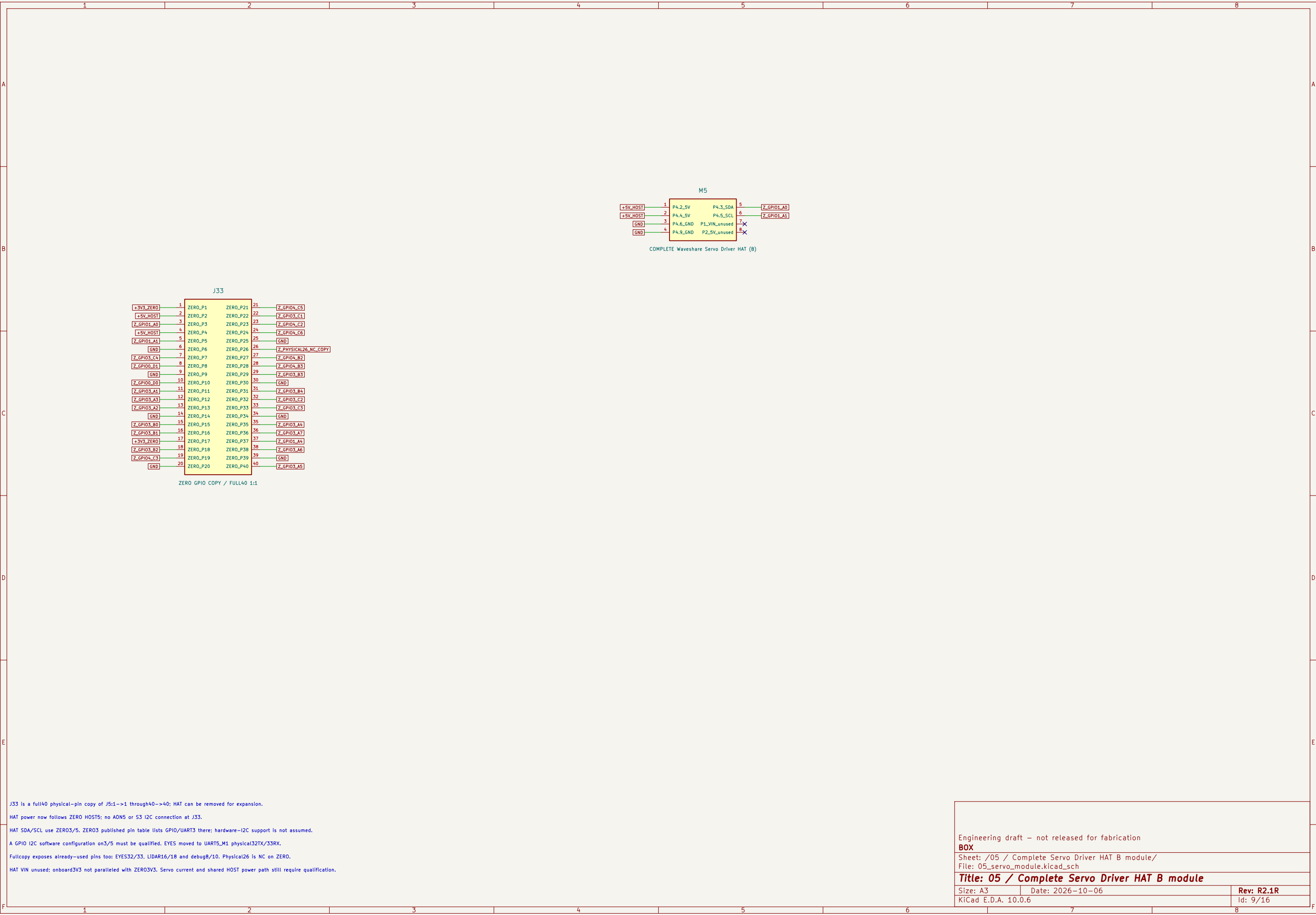
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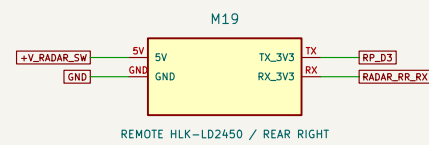
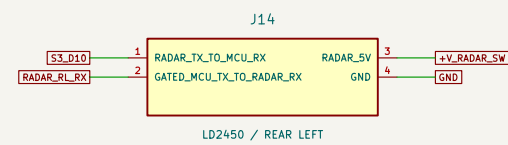
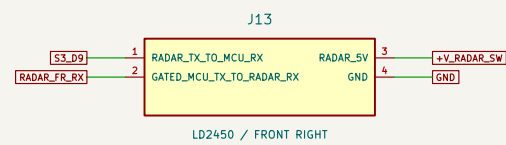
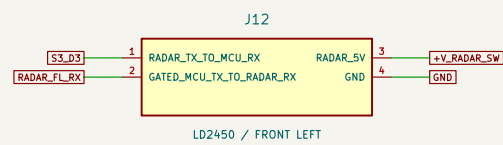
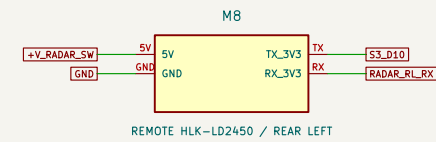
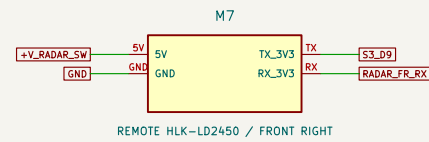
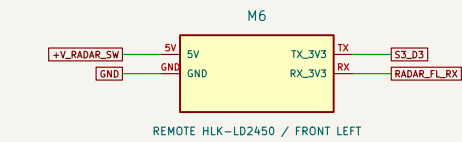
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Four remote HLK-LD2450 modules use J12/J13/J14/J36 Grove harness sockets.

All four radar sockets:1=radarTX/hostRX,2=radarRX/hostTX,3=switched5V,4=GND.

HLK-LD2450 manufacturer specifies 5V supply and 3.3V TTL UART. Off-power tolerance is unverified.

Use the actual HLK sensor cable end; map wires by signal, not by connector-shell resemblance.

M2 NC inhibits both existing muxes when radar power is off. Release TX before either relay transition.

Engineering draft – not released for fabrication

BOX

Sheet: /06 / Four remote HLK-LD2450 harnesses/
File: 06_radar_harness.kicad_sch

Title: 06 / Four remote HLK-LD2450 harnesses

Size: A3

Date: 2026-10-06

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Id: 10/16



Existing ReSpeaker XVF3800 + XIAO retained; audio/CDC cable uses XIAO native USB.

This sheet documents off-carrier USB endpoints, not new modules or copper USB traces on the carrier.

User requires USB-only connection. No J26 GPIO/power harness and no J22 shutdown GPIO Interface.

ReSpeaker USB is powered by ZERO HOST VBUS. Wake requires the always-on controller/panel; an off ZERO cannot receive USB commands.

No XVF3800 GPIO harness selected. USB cable/adaptor power behavior remains a physical qualification item.

Engineering draft – not released for fabrication

BOX

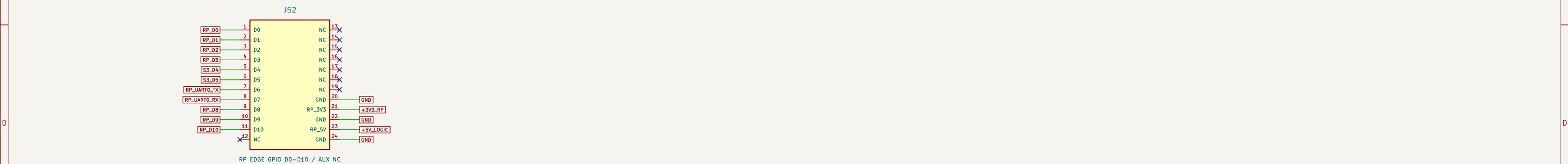
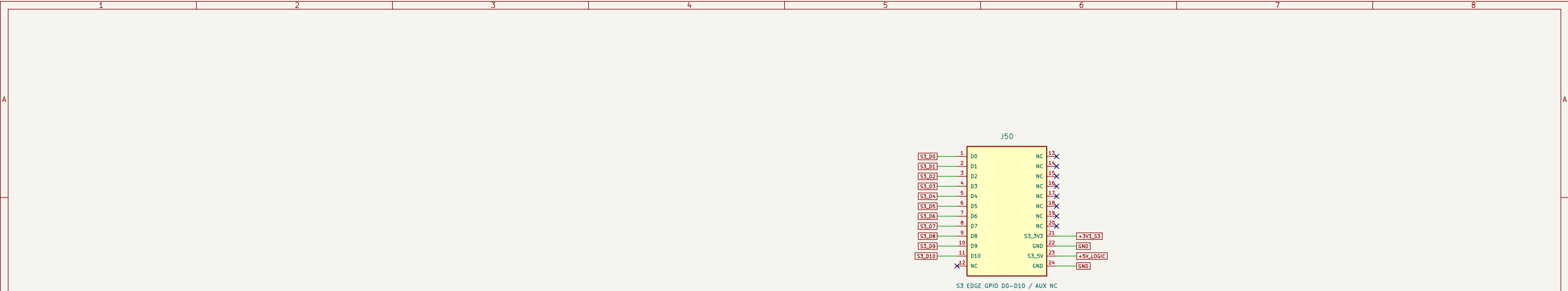
Sheet: /10 / Existing ReSpeaker USB link/
File: 10_respeaker_usb.kicad_sch

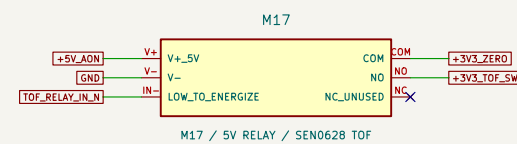
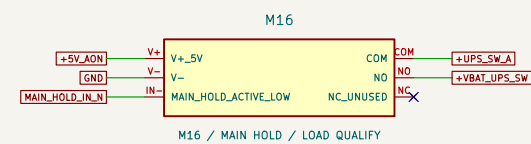
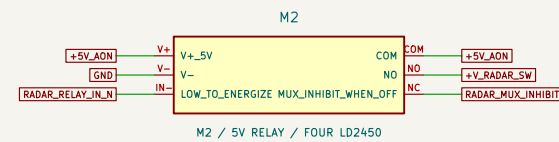
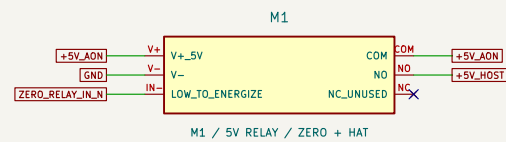
Title: 10 / Existing ReSpeaker USB link

Size: A3Date: 2026-10-06KiCad E.D.A. 10.0.6

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[illegible]



ALL FOUR ORIGINAL RELAYS retained with original COM/NO load circuits and LOW=ON.

M1 ZERO/HAT: RP_D1. M2 RADAR: S3_D1. M16 MAIN: RP_D2. M17 TOF: RP_D8.

J51 START still bridges UPS battery switch loop. NEITHER PIN IS GND.

Hold START until RP firmware asserts MAIN LOW; orderly shutdown releases MAIN last.

HFD4 photo marking: manufacturer contact rating 1A/30VDC resistive, max switching 2A; actual module load/inrush unqualified.

Engineering draft – not released for fabrication

BOX

Sheet: /12 / Three load relays and whole-device power hold/

File: 12_four_power_relays.kicad_sch

Title: 12 / Three load relays and whole-device power hold

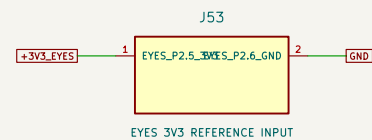
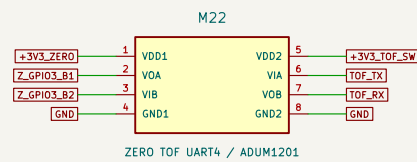
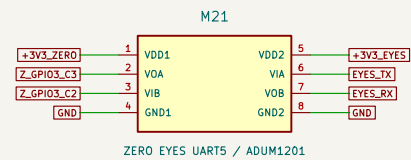
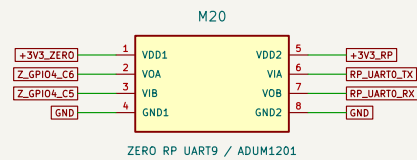
Size: A3

Date: 2026-10-06

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M20 RP/ZERO, M21 EYES/ZERO, M22 ZERO/TOF. All three are complete LA161062 assemblies.

Each isolator side uses its actual endpoint3V3. Domain separation is implemented; brownout behavior needs bench validation.

J53 is EYES P2.5/P2.6 reference INPUT; never bridge it to RP or ZERO 3V3.

Common GND is intentional. These modules separate logic power domains, not safety isolation.

Carrier rear solder lands are nominal; use short tinned wire contacts, no sockets or tall headers.

USB/battery bench service: remove endpoint harnesses; XIAOs are removed from carrier for USB.

Engineering draft – not released for fabrication

BOX

Sheet: /13 / Rear complete UART modules/

File: 13_power_off_uart.kicad_sch

Title: 13 / Rear complete UART modules

Size: A3

Date: 2026-10-06

Rev: R2.1R

KiCad E.D.A. 10.0.6

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FAN PWM / POLOLU 2990

TACH remains ZERO GPIO3_A1; enable only its ZERO 3V3 pull-up.

Id: 16/16